

PES UNIVERSITY – DEPARTMENT OF CSE

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #50 | Developer Tools & IT Operations

Centralized Audit Trail Compliance Engine

Problem context: A compliance logging platform that ingests enterprise application event logs, masks sensitive PII fields in accordance with GDPR, and generates cryptographically verifiable immutable audit exports.

1. Complete Requirements Table

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest enterprise application audit logs in structured JSON format.	High	Pass: a valid JSON audit event is accepted and made available for processing. Fail: malformed or unsupported input is persisted as a valid audit record.	Reliable ingestion is required to centralize audit evidence.
FR-002	Functional	The system shall parse structured JSON audit logs and identify sensitive PII fields, including Social Security Numbers and credit-card numbers.	High	Pass: all configured PII fields in a test log are detected. Fail: a known PII field remains unidentified.	PII identification is the prerequisite for GDPR-compliant masking.
FR-003	Functional	The system shall mask identified PII fields before persisting audit records to storage.	High	Pass: the persisted record contains masked PII and no cleartext SSN or credit-card number. Fail: any protected field is stored in cleartext.	Prevents sensitive information from being retained in unprotected form.
FR-004	Functional	The system shall persist protected audit records together with a SHA-256 integrity hash.	High	Pass: every persisted audit record has a corresponding SHA-256 hash and the hash validates against the protected record. Fail: a record is stored without a valid integrity hash.	Supports tamper-evidence and verifiable audit history.
FR-005	Functional	The system shall allow authorized stakeholders to query and filter historical audit records and generate immutable audit exports.	High	Pass: an authorized query returns matching records and an export can be generated with verifiable integrity information. Fail: the system returns unauthorized data or an unverifiable export.	Enables compliance review and production of trustworthy audit evidence.
NFR-001	Performance & Security	Audit-log query endpoints shall support complex filtering across 10 million historical records in under 1 second.	High	Pass: benchmark tests under simulated peak load demonstrate complex queries completing in less than 1 second while meeting security controls.	Fast historical retrieval is required for practical compliance and security investigations.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Security & Compliance	The platform shall protect PII according to GDPR requirements and ensure audit exports are cryptographically verifiable and immutable.	High	Pass: test cases confirm PII is masked before persistence and generated exports cannot be modified without integrity verification detecting the change.	Protects sensitive data and preserves the evidentiary value of audit records.

2. UML Use-Case Diagram

Actors: Compliance Officer and Security Auditor. The diagram models the primary use cases and includes both «include» and «extend» relationships.

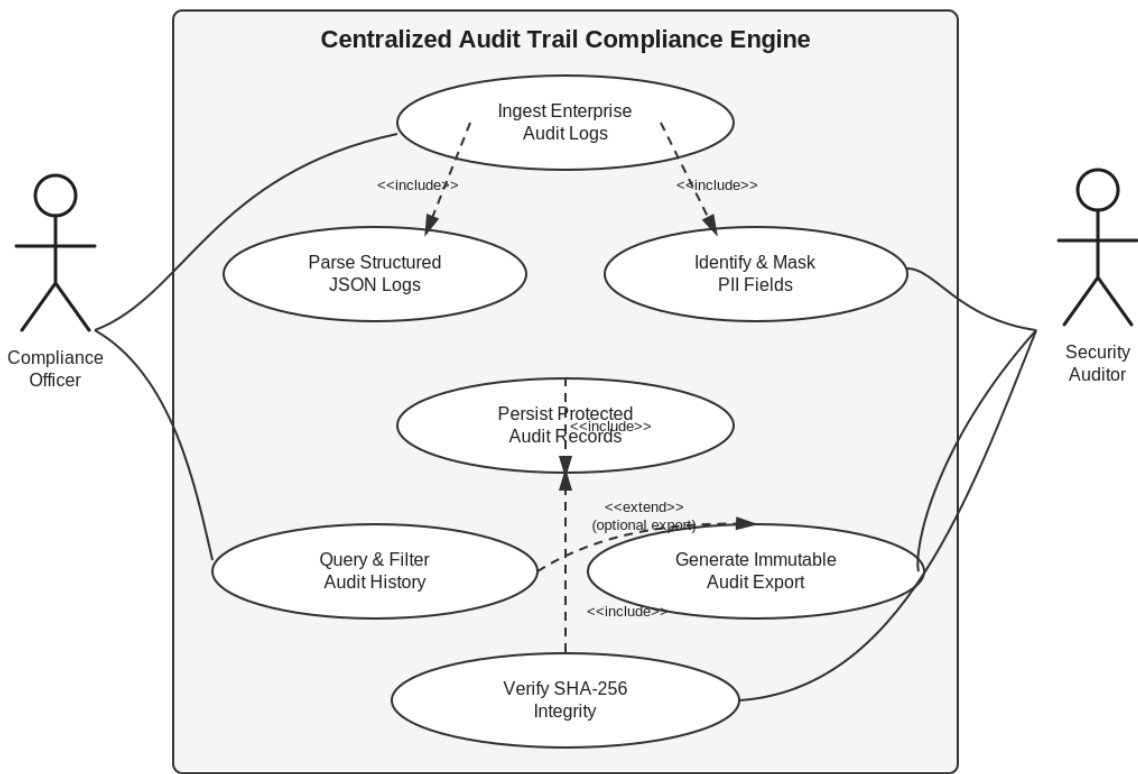


Figure 1. Draw.io-style UML use-case diagram for the Centralized Audit Trail Compliance Engine.

3. Use-Case Flow Specification

Core Use Case: Generate Immutable Audit Export

Preconditions

- Audit records have been ingested, parsed, PII-masked, and persisted.
- Persisted records have associated SHA-256 integrity hashes.
- The Security Auditor or Compliance Officer is authenticated and authorized to access the required audit scope.

Postconditions

- A requested audit dataset is exported in an immutable format.

- The export contains integrity information that can be cryptographically verified.
- The export is available to the authorized stakeholder without exposing cleartext protected PII.

Main Success Scenario

1. The actor selects the required historical audit scope and applies filters.
2. The system authenticates and authorizes the request.
3. The system executes the requested filters over historical audit records.
4. The system retrieves the matching protected records.
5. The system generates an immutable audit export from the retrieved records.
6. The system associates/verifies SHA-256 integrity information for the export.
7. The system presents the completed export to the authorized actor.

Alternate Flow

- A1. If integrity verification fails for any record or export component, the system stops the export, marks the verification as failed, and reports the integrity failure to the actor instead of presenting the export as valid.

Source basis: Problem Statement #50 provided for Lab 1; the requirements above expand the supplied functional and non-functional guidelines into the exact required 5 FR + 2 NFR deliverables.